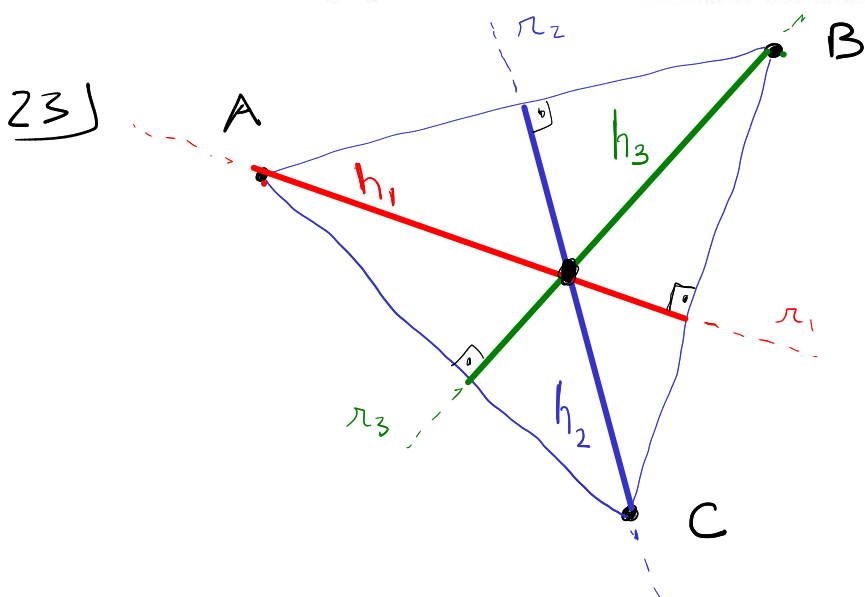


23 Determine as equações das alturas do triângulo de vértices $A(-3, 2)$, $B(5, 4)$, $C(3, -8)$ e ache o ponto de sua interseção.

24 Determine as equações das medianas do triângulo do Exercício 23 e ache o ponto de interseção delas.



$\overline{BC}$:

$$m_{\overline{BC}} = \frac{y_B - y_C}{x_B - x_C} = \frac{4 - (-8)}{5 - 3} = \frac{12}{2} = 6$$

$h_1 \perp \overline{BC}$:

$$m_{h_1} \cdot m_{\overline{BC}} = -1 \quad (\text{TEOREMA})$$

$$m_{h_1} \cdot 6 = -1 \Rightarrow m_{h_1} = -\frac{1}{6}$$

Reta r_1 :

$$m = -\frac{1}{6}$$

$A \in r_1$

(x_1, y_1)

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{1}{6}(x - (-3))$$

$$y - 2 = -\frac{1}{6}(x + 3)$$

reta 2

$h_3 \perp \overline{AC}$

$$m_{\overline{AC}} = \frac{y_C - y_A}{x_C - x_A} = \frac{-8 - 2}{3 - (-3)} = \frac{-10}{6} = -\frac{5}{3}$$

$$m_{h_3} \cdot m_{\overline{AC}} = -1 \Rightarrow m_{h_3} = \frac{3}{5}$$

P/reta 3:

$$m_{h_3} = \frac{3}{5}$$

$B \in r_3$

(x_1, y_1)

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \frac{3}{5}(x - 5)$$

reta 3

$$h_2 \perp \overline{AB} \Rightarrow m_{h_2} \cdot m_{AB} = -1$$

$$m_{h_2} \cdot \frac{2}{8} = -1 \Rightarrow m_{h_2} = -4$$

P/r₂:

$$\left. \begin{array}{l} m = -4 \\ C \in r_2 \\ (x_1, y_1) \end{array} \right\} \Rightarrow \begin{array}{l} y - y_1 = m(x - x_1) \\ y - (-8) = -4(x - 3) \\ \boxed{y + 8 = -4(x - 3)} \\ \text{reta } r_2 \end{array}$$

CONSIDERANDO r_2 e r_3 : *Buscamos $r_2 \cap r_3$*

$$r_2: y + 8 = -4(x - 3) \Leftrightarrow y + 8 = -4x + 12 \quad (x-1)$$

$$r_3: y - 4 = \frac{3}{5}(x - 5) \Leftrightarrow y - 4 = \frac{3}{5}x - 3$$

$$\Leftrightarrow \begin{cases} -y - 8 = 4x - 12 \\ y - 4 = \frac{3}{5}x - 3 \end{cases} + \quad y + \frac{3}{5} = \frac{20}{5} + \frac{3}{5} = \frac{23}{5}$$

$$-12 = \frac{23}{5}x - 15 \Rightarrow \frac{23}{5}x = 3 \Rightarrow \boxed{x = \frac{15}{23}}$$

Substituindo em r_2 :

$$x = \frac{15}{23} \Rightarrow y + 8 = -4 \left(\frac{15}{23} - 3 \right)$$

$$y + 8 = -4 \left(-\frac{54}{23} \right)$$

$$\frac{15}{23} - \frac{3}{1} = \frac{15 - 69}{23}$$

$$= -\frac{54}{23}$$

$$y + 8 = \frac{216}{23} \Rightarrow y = \frac{216}{23} - 8$$

$$\begin{array}{r} 23 \\ \times 8 \\ \hline 184 \end{array}$$

$$y = \frac{216 - 184}{23} = \frac{32}{23}$$

Ou seja, $\left(\frac{15}{23}, \frac{32}{23} \right) = r_2 \cap r_3$

Será que $(\frac{15}{23}, \frac{32}{23}) \in \pi_2$?

TESTE

$$y - 2 = -\frac{1}{6}(x + 3)$$

$$\frac{32}{23} - 2 = -\frac{1}{6}\left(\frac{15}{23} + 3\right)$$

$$\frac{15}{23} + \frac{69}{23} = \frac{84}{23}$$

$$\frac{32}{23} - 2 = -\frac{1}{6}\left(\frac{84}{23}\right)$$

$$\frac{32 - 46}{23} = -\frac{1}{6}\left(\frac{84}{23}\right) \Rightarrow -14 = -\frac{1}{6}(84)$$

VERDADE

$$-84 = -84$$

OU SEJA: $(\frac{15}{23}, \frac{32}{23}) \in \pi_1$

Logo, $(\frac{15}{23}, \frac{32}{23}) \in \pi_1 \cap \pi_2 \cap \pi_3$

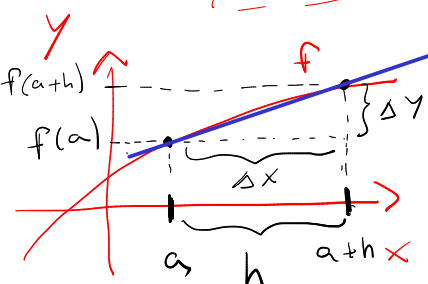
Seção 1.4

IMPORTANT E
P/ DERIVADA

4.f)

$$\frac{f(a+h) - f(a)}{h}$$

sendo $f(x) = \frac{1}{x^2 + 1}$



reta secante
ao gráfico
de f por

$(a, f(a))$ e $(a+h, f(a+h))$

Qual é o coef. angular? $m = \frac{f(a+h) - f(a)}{h}$

$$\left. \begin{aligned} f(a+h) &= \frac{1}{(a+h)^2+1} \\ f(a) &= \frac{1}{a^2+1} \end{aligned} \right\} f(a+h) - f(a) = \frac{1}{(a+h)^2+1} - \frac{1}{a^2+1}$$

$$= \frac{1}{\overset{\times a^2+1}{a^2+1}} - \frac{1}{\overset{\times a^2+2ah+h^2+1}{a^2+1}}$$

$$= \frac{\cancel{a^2+1} - (\cancel{a^2+2ah+h^2+1})}{(a^2+1)(a^2+2ah+h^2+1)}$$

$$= \frac{-2ah - h^2}{(a^2+1)(a^2+2ah+h^2+1)} \Bigg\} = f(a+h) - f(a)$$

$$\frac{f(a+h) - f(a)}{h} = \frac{1}{h} \cdot (f(a+h) - f(a))$$

$$= \frac{1}{\cancel{h}} \cdot \frac{\overbrace{h(-2a-h)}^{h(-2a-h)} \quad \cancel{(-2ah-h^2)}}{(a^2+1)(a^2+2ah+h^2+1)}$$

$$= \frac{-2a - h}{(a^2+1)(a^2+2ah+h^2+1)}$$

OBS

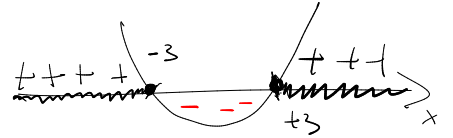
$$\lim_{h \rightarrow 0} \frac{-2a - 0}{(a^2+1)(\cancel{a^2+2a \cdot 0} + \cancel{0^2}+1)} = \frac{-2a}{(a^2+1)(a^2+1)} = \frac{-2a}{(a^2+1)^2}$$

COEF. Angular da reta tangente ao graf. f pelo ponto (a, f(a))

ISSO É A DERIVADA DE f em x=a

$$10. f(x) = \sqrt{\underbrace{x^2 - 9}_{\geq 0}}$$

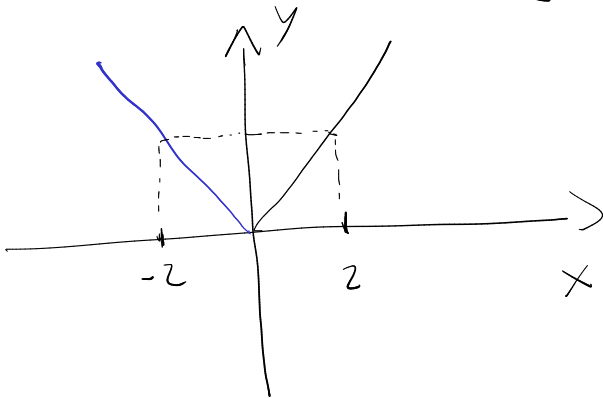
$$x^2 - 9 \geq 0$$



$$\begin{aligned} D_f &= (-\infty, -3] \cup [3, +\infty) \\ &= \mathbb{R} - (-3, 3) \end{aligned}$$

$$f: \mathbb{R} - (-3, 3) \longrightarrow \mathbb{R}_+$$

$$25) f(x) = |x| : \begin{cases} x, & \text{se } x \geq 0 \\ -x, & \text{se } x < 0 \end{cases}$$



$$D_f = \mathbb{R}$$

$$Im f = \mathbb{R}_+$$

$$\begin{aligned} f: \mathbb{R} &\longrightarrow \mathbb{R}_+ \\ x &\longmapsto |x| \end{aligned}$$

É sobrejetora.
Pois:
 $a \in \mathbb{R}_+ \Rightarrow$ existe
 $x \in D_f$ tal que
 $f(x) = a$.
Basta tomar
 $x = a$.

NÃO é injetora.

Exemplo:

$$-2 \neq 2 \wedge \{-2, 2\} \subset D_f$$

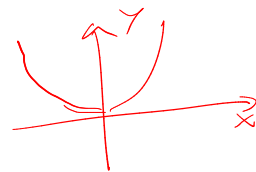
$$\text{Porém, } f(-2) = f(2)$$

→ Não é injetora

27]

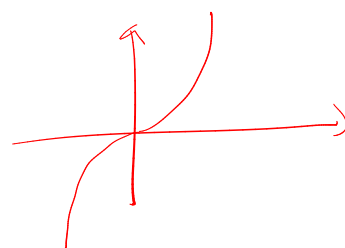
Função : $f(x) = f(-x)$, para todo $x \in D_f$
PAR

↓
SIMETRIA
COM EIXO Y → Ex: $f(x) = x^2$



Função : $f(x) = -f(-x)$, para todo $x \in D_f$
ÍMPAR

↓
SIMETRIA
COM A ORIGEM
(0,0) → Ex: $f(x) = x^3$



$$f(x) = 3x^3 - 4x$$

$$f(-x) = 3(-x)^3 - 4(-x) = -3x^3 + 4x \neq f(x) \rightarrow \text{Não é f. par}$$

Mas

$$-1 \times f(-x) = f(x) \rightarrow \text{É função ÍMPAR}$$